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LEARNER GUIDE

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Introduction

This Learner Guide accompanies the course AWS Certified AI Practitioner Training (AIF-C01), conducted by Tertiary Infotech Academy Pte Ltd. It is your hands-on companion to the slide deck: the deck teaches the concepts, and this guide turns each concept into a short lab you complete in your own AWS account. The labs are grouped under the five official AIF-C01 exam domains and run in order, so the skills accumulate toward both the exam and the practical assessment. Use each lab's Objective and Goal to see what you are building and why it matters, follow the numbered Step-by-step, and keep every output as revision material. The Slides line in each lab points back to the deck for the underlying theory.

Course Learning Outcomes

- LO1 — Implement AI applications with Amazon Q Business and AWS AI models.
- LO2 — Deploy AWS AI Bedrock workflows according to plan and machine learning techniques.
- LO3 — Identify and report issues with AI applications, data and implementation procedures.
- LO4 — Maintain data interoperability during AI development per data-management principles.
- LO5 — Perform data cleaning techniques to optimise AWS foundation models.

Before You Start

What you need

- An AWS account — a free-tier account is enough: <https://aws.amazon.com/free/>
- The AWS CLI installed and configured (run `aws configure`, then `aws sts get-caller-identity` to confirm).
- The course slides and this Learner Guide.

Secure your account first

- Enable MFA on the root user.
- Create an IAM admin user and stop using root for daily work.
- Set a billing alarm so runaway resources surface early.

Set your working Region and conveniences

- Work in `us-east-1` unless a lab says otherwise — AI/ML features appear there first.
- Set `AWS_DEFAULT_REGION=us-east-1` and `AWS_PAGER=""` for a smoother CLI experience.

Amazon Bedrock model access

- Labs 06 onward need foundation model access — request it before class.
- In the Bedrock console, choose Model access, then request the models you need.
- Approval is not always instant, and available models vary by Region.

Cost awareness

- Most labs stay within the free tier, but several create resources that bill until deleted.
- Billing resources appear in Labs 04, 05, 20, 21, 22 (SageMaker), 15, 16 (OpenSearch Serverless), 11 (Amazon Q Business) and 24 (Macie). Do not provision throughput in Lab 17.
- Always complete each lab's Clean up step, and verify OpenSearch collections in every Region with `aws opensearchserverless list-collections`.

Verify your setup

- `aws sts get-caller-identity` — returns your account and identity.
- `aws bedrock list-foundation-models` — returns the models available in your Region.

DOMAIN 1 — Fundamentals of AI and ML (20%)

Lab 1 — AI, ML and Deep Learning Terminology [Demo]

Objective: Build a precise vocabulary for AI, ML, deep learning and generative AI and locate each in the AWS console.

Goal: These four terms nest inside one another; using them precisely is the foundation the exam and every later lab build on.

What you'll build

A layered AI / ML / deep-learning / generative-AI concept map tied to the AWS services that embody each layer.

At a glance — Domain 1, Duration 30 min, Slides 25–51.

Step-by-step

1. Survey the AWS AI and ML console landscape
2. Map the AI, ML, deep learning and generative AI hierarchy
3. Distinguish supervised, unsupervised and reinforcement learning
4. Separate algorithm, model, training and inference
5. Compare inferencing options: real-time versus batch
6. Clean up

Lab 2 — Data Types and Labelling for ML [Demo]

Objective: Classify structured, semi-structured and unstructured data and produce a labelled image dataset.

Goal: Model quality starts with data; knowing data types and labelled versus unlabelled data is prerequisite to any ML work.

What you'll build

An S3-hosted dataset plus a SageMaker Ground Truth labelling job and its output manifest.

At a glance — Domain 1, Duration 35 min, Slides 25–51.

Step-by-step

7. Classify structured, semi-structured and unstructured data
8. Create an S3 bucket and organise a dataset
9. Understand labelled versus unlabelled data
10. Create a SageMaker Ground Truth labelling job
11. Label the images and read the output manifest
12. Clean up

Lab 3 — AWS Managed AI Services Tour [Demo]

Objective: Use Amazon Comprehend, Translate, Polly, Transcribe and Rekognition from the console.

Goal: AWS managed AI services solve common tasks with no model training; matching each to a use case is core exam knowledge.

What you'll build

A worked example in each of the five managed AI services and a service-to-use-case matching table.

At a glance — Domain 1, Duration 40 min, Slides 52–83.

Step-by-step

13. Extract meaning from text with Amazon Comprehend
14. Translate text with Amazon Translate
15. Synthesise speech with Amazon Polly
16. Convert speech to text with Amazon Transcribe
17. Analyse images with Amazon Rekognition
18. Match services to use cases
19. Clean up

Lab 4 — Train and Deploy a Model with SageMaker [Self-paced]

Objective: Train, deploy and evaluate a binary classifier with SageMaker built-in XGBoost, then delete every billing resource.

Goal: Walking the full train-deploy-infer loop makes the ML lifecycle concrete and shows where cost accrues.

What you'll build

A trained XGBoost model served from a real-time SageMaker endpoint.

At a glance — Domain 1, Duration 45 min, Slides 84–135.

Step-by-step

20. Read the cost warning and plan the run
21. Create a SageMaker notebook instance
22. Prepare the dataset and upload it to S3
23. Train a model with the built-in XGBoost algorithm
24. Deploy the model to a real-time endpoint
25. Run inference and evaluate the model
26. Clean up

Lab 5 — The ML Pipeline and Data Preparation [Self-paced]

Objective: Design a complete ML pipeline for a churn scenario, from business framing to monitoring.

Goal: Every production model follows the same pipeline; being able to place each stage is essential to the exam and to real work.

What you'll build

An end-to-end ML pipeline design grounded in the SageMaker console, from data prep and feature engineering to deployment and monitoring.

At a glance — Domain 1, Duration 40 min, Slides 84–135.

Step-by-step

27. Frame the business problem and map the pipeline
28. Collect data and explore it
29. Pre-process the data
30. Engineer features and store them
31. Train, tune and evaluate
32. Deploy, monitor and close the loop
33. Clean up

DOMAIN 2 — Fundamentals of Generative AI (24%)

Lab 6 — Amazon Bedrock Getting Started [Demo]

Objective: Enable foundation model access in Amazon Bedrock and run your first prompt.

Goal: Bedrock is the gateway to foundation models on AWS; model access and the playground underpin every generative-AI lab that follows.

What you'll build

An access-enabled Bedrock account, a tour of the model catalogue and a first playground prompt.

At a glance — Domain 2, Duration 30 min, Slides 137–164.

Step-by-step

34. Open Amazon Bedrock and choose a Region
35. Request access to foundation models
36. Explore the model catalogue
37. Run your first prompt in the playground
38. See the same catalogue from the AWS CLI
39. Clean up

Lab 7 — Comparing Foundation Models and Inference Parameters [Demo]

Objective: Compare foundation models on one prompt, then vary temperature, top-P and max tokens.

Goal: Model choice and inference parameters shape every generative output and are directly tested.

What you'll build

A side-by-side model comparison and a record of how each inference parameter changes the output.

At a glance — Domain 2, Duration 35 min, Slides 137–164.

Step-by-step

- 40. Set up the comparison and run the first model
- 41. Run the identical prompt on a second model
- 42. Baseline at low temperature
- 43. Raise temperature, then narrow top-P
- 44. Max tokens truncates, it does not shorten
- 45. Clean up

Lab 8 — Tokens, Embeddings and Vector Representations [Demo]

Objective: Turn text into tokens, generate embeddings and measure semantic similarity.

Goal: Tokens drive cost and context limits, and embeddings power search and RAG; both are recurring exam concepts.

What you'll build

A token-count analysis and embedding vectors compared for semantic similarity.

At a glance — Domain 2, Duration 35 min, Slides 137–164.

Step-by-step

- 46. Tokens are not words
- 47. Tokens drive cost and the context window
- 48. Generate an embedding
- 49. Semantic similarity — compare the vectors
- 50. Where this leads — vector stores and RAG
- 51. Clean up

Lab 9 — Generative AI Capabilities and Limitations [Demo]

Objective: Exercise what generative AI does well, produce a hallucination, and apply mitigations.

Goal: Knowing both the capabilities and the limitations of generative AI is a whole exam task statement.

What you'll build

A documented hallucination example with grounding and human-review mitigations applied.

At a glance — Domain 2, Duration 30 min, Slides 165–172.

Step-by-step

- 52. Capabilities — summarisation and generation
- 53. Capabilities — translation and code

- 54. Produce a hallucination yourself and record it
- 55. The other four limitations
- 56. Mitigations — grounding, review and scoping
- 57. Clean up

Lab 10 — Choosing Bedrock, SageMaker JumpStart or a Managed Service

[Demo]

Objective: Compare Bedrock, SageMaker JumpStart and managed services and classify ten scenarios.

Goal: Choosing the right AWS AI layer for a task is a frequent exam decision.

What you'll build

A decision table and ten business scenarios classified against it.

At a glance — Domain 2, Duration 35 min, Slides 173–182.

Step-by-step

- 58. The three options and the question that separates them
- 59. Option A — a managed AI service
- 60. Option B — Bedrock, and Option C — SageMaker JumpStart
- 61. Build the decision table
- 62. Classify ten scenarios
- 63. Clean up

Lab 11 — Amazon Q Business Getting Started [Self-paced]

Objective: Create an Amazon Q Business application answering questions from your own documents with citations.

Goal: Amazon Q Business shows managed, grounded generative AI in action over enterprise content.

What you'll build

A Q Business application connected to S3 documents, returning cited answers.

At a glance — Domain 2, Duration 40 min, Slides 173–182.

Step-by-step

- 64. What Amazon Q Business is and what it will cost
- 65. Prepare documents in Amazon S3
- 66. Create the Amazon Q Business application
- 67. Connect the S3 data source and sync
- 68. Ask questions grounded in your documents
- 69. Test the boundaries of grounding
- 70. Clean up

DOMAIN 3 — Applications of Foundation Models (28%)

Lab 12 — Prompt Engineering Basics — Zero-Shot and Few-Shot [Demo]

Objective: Run one task at zero-shot, one-shot and few-shot and record how quality changes.

Goal: Prompt engineering is the largest exam domain and shot count is its most basic lever.

What you'll build

The same classification task at three shot levels with a quality comparison.

At a glance — Domain 3, Duration 35 min, Slides 210–235.

Step-by-step

71. Set up the playground and fix one task
72. The anatomy of a prompt
73. Zero-shot prompting
74. One-shot and few-shot prompting
75. Compare, and find where more examples stop helping
76. Clean up

Lab 13 — Advanced Prompting and Prompt Templates [Demo]

Objective: Apply system prompts, roles, chain-of-thought and negative prompting, then package a template.

Goal: Advanced prompting techniques and reusable templates are core to applying foundation models.

What you'll build

A tested, reusable prompt template for a rule-heavy triage task.

At a glance — Domain 3, Duration 40 min, Slides 210–235.

Step-by-step

77. Set up and define the scenario
78. System prompts and role assignment
79. Chain-of-thought prompting
80. Negative prompting and constraints
81. Build a reusable prompt template
82. Test the template against edge cases
83. Clean up

Lab 14 — Prompt Misuse, Injection and Defences [Demo]

Objective: Probe your own triage application for prompt risks and build a five-layer defence.

Goal: Prompt injection and misuse are named exam risks; layered defence is the expected mitigation.

What you'll build

A threat model plus a five-layer defence covering prompt hardening, input validation, output filtering, Guardrails and least privilege.

At a glance — Domain 3, Duration 35 min, Slides 210–235.

Step-by-step

- 84. The application you own, and its threat model
- 85. The four risk classes
- 86. Probe your own application
- 87. Defence layers 1 and 2 — prompt hardening and input validation
- 88. Defence layers 3 and 4 — output filtering and Guardrails
- 89. Clean up

Lab 15 — RAG with Knowledge Bases for Amazon Bedrock [Self-paced]

Objective: Build a RAG pipeline over your own documents with Knowledge Bases for Amazon Bedrock.

Goal: RAG is the primary way to ground a model in private data and a heavily tested concept.

What you'll build

A Bedrock knowledge base returning cited answers, proven against an ungrounded baseline.

At a glance — Domain 3, Duration 45 min, Slides 184–209.

Step-by-step

- 90. Cost warning, and prepare the S3 data source
- 91. How RAG works
- 92. Create the knowledge base
- 93. Sync the data source
- 94. Retrieve and generate, and read the citations
- 95. Compare against the no-context baseline
- 96. Clean up

Lab 16 — Vector Stores and Retrieval Quality [Self-paced]

Objective: Tune chunk size, overlap, top-K and metadata filtering, and diagnose retrieval failures.

Goal: Retrieval quality decides RAG quality; telling a retrieval failure from a generation failure is a practical skill.

What you'll build

A tuned retrieval configuration and a diagnosis of a bad answer.

At a glance — Domain 3, Duration 40 min, Slides 184–209.

Step-by-step

- 97. Set up and measure a baseline
- 98. Chunk size and overlap
- 99. Top-K

100. Metadata filtering
101. Diagnose a bad answer
102. Clean up

Lab 17 — Fine-Tuning and Model Customisation [Self-paced]

Objective: Prepare a fine-tuning dataset and configure a Bedrock custom model job without submitting it.

Goal: Knowing when to fine-tune versus prompt, RAG or continue pre-training is an exam decision with real cost consequences.

What you'll build

A fine-tuning dataset, a configured custom-model job (not submitted) and a customisation decision table.

At a glance — Domain 3, Duration 40 min, Slides 236–247.

Step-by-step

103. The customisation decision
104. Prepare a fine-tuning dataset
105. Configure a custom model job — do not submit
106. Provisioned throughput, and why you stop here
107. Apply the decision to real scenarios
108. Clean up

Lab 18 — Evaluating Foundation Model Performance [Self-paced]

Objective: Run a Bedrock automatic model evaluation and place metrics in a full evaluation strategy.

Goal: Evaluating model performance across automatic metrics, human review and business metrics is a distinct exam task.

What you'll build

An automatic evaluation job result interpreted alongside ROUGE, BLEU, BERTScore, benchmarks and business metrics.

At a glance — Domain 3, Duration 40 min, Slides 248–271.

Step-by-step

109. Why evaluate, and build a test set
110. The automatic metrics
111. Run an automatic evaluation job
112. Interpret the results
113. Human evaluation, benchmarks, and business metrics
114. Clean up

DOMAIN 4 — Guidelines for Responsible AI (14%)

Lab 19 — Amazon Bedrock Guardrails [Demo]

Objective: Create a Bedrock guardrail with content, denied-topic, word and PII filters and test it.

Goal: Guardrails are the AWS control for responsible generative AI and are directly examinable.

What you'll build

A configured Bedrock guardrail with a record of which policy blocks each test prompt.

At a glance — Domain 4, Duration 40 min, Slides 273–284.

Step-by-step

115. Create the guardrail and set blocked messages
116. Configure content filters
117. Configure denied topics
118. Configure word filters
119. Configure sensitive information filters (PII)
120. Test the guardrail and record findings
121. Clean up

Lab 20 — Bias Detection with SageMaker Clarify [Self-paced]

Objective: Frame a fairness question and compare pre- and post-training bias metrics with SageMaker Clarify.

Goal: Detecting bias is a responsible-AI requirement the exam expects you to reason about.

What you'll build

A fairness framing and a pre / post-training bias comparison from a Clarify job or guided report.

At a glance — Domain 4, Duration 40 min, Slides 273–284.

Step-by-step

122. Frame the fairness question: facet, label and positive outcome
123. Pre-training bias metrics: measuring the data
124. Post-training bias metrics: measuring the model
125. Path 1: run a Clarify bias job against your own model
126. Path 2: guided walkthrough of a bias report
127. Clean up

Lab 21 — Explainability and Feature Attribution [Demo]

Objective: Produce global and local SHAP explanations and decide when interpretability must win.

Goal: Explainability underpins trust in AI and is a responsible-AI exam topic.

What you'll build

A Clarify SHAP feature-attribution report with a performance-versus-interpretability decision.

At a glance — Domain 4, Duration 35 min, Slides 285–297.

Step-by-step

128. Global versus local explanations
129. SHAP feature attribution and the baseline
130. Configure a Clarify explainability analysis
131. Read and interpret a feature attribution report
132. Performance versus interpretability, and when to reject the better model
133. Clean up

Lab 22 — Model Cards and Transparency [Demo]

Objective: Create a SageMaker model card and distinguish it from Model Monitor and the Model Registry.

Goal: Model cards document intended use and risk; they are the transparency artefacts the exam names.

What you'll build

A completed SageMaker model card, compared with AWS AI Service Cards.

At a glance — Domain 4, Duration 30 min, Slides 285–297.

Step-by-step

134. Create a model card and declare intended use
135. Set the risk rating and record training details
136. Record evaluation results and caveats
137. Model cards versus Model Monitor versus Model Registry
138. AWS AI Service Cards and the transparency you inherit
139. Clean up

DOMAIN 5 — Security, Compliance and Governance for AI Solutions (14%)

Lab 23 — Securing AI Workloads with IAM and KMS [Self-paced]

Objective: Secure an AI workload with least-privilege IAM, customer managed KMS keys and VPC endpoints.

Goal: Securing AI workloads and knowing the shared-responsibility boundary is a whole exam domain.

What you'll build

A scoped IAM policy, a KMS-encrypted data bucket and private VPC-endpoint access to Bedrock.

At a glance — Domain 5, Duration 40 min, Slides 299–333.

Step-by-step

140. Map the shared responsibility model onto an AI workload
141. Write a least-privilege IAM policy scoped to specific Bedrock models
142. Create a KMS customer managed key for AI data
143. Encrypt a training data bucket at rest, and understand encryption in transit
144. Keep AI traffic off the public internet with VPC endpoints
145. Clean up

Lab 24 — Data Protection with Amazon Macie [Self-paced]

Objective: Use Amazon Macie to discover synthetic personal data in S3 and turn findings into controls.

Goal: Keeping PII out of training data and prompts is a governance requirement Macie enforces.

What you'll build

A Macie discovery job, its findings, and the data controls they drive.

At a glance — Domain 5, Duration 35 min, Slides 299–333.

Step-by-step

146. Why PII must never reach training data or prompts
147. Create a bucket and upload synthetic sample data
148. Enable Amazon Macie
149. Create a custom data identifier and run a discovery job
150. Read the findings and turn them into data controls
151. Clean up

Lab 25 — Governance, Audit and Model Lineage [Demo]

Objective: Inspect CloudTrail events, add AWS Config compliance rules and use the SageMaker Model Registry.

Goal: Audit, compliance and model lineage are the governance controls that close the AI lifecycle.

What you'll build

A CloudTrail audit trail, AWS Config compliance rules and a Model Registry approval workflow.

At a glance — Domain 5, Duration 35 min, Slides 334–374.

Step-by-step

152. Generate AI API activity to audit
153. Inspect a real CloudTrail event: identity, action, timestamp
154. Create a trail for long-term audit retention
155. Continuous compliance checking with AWS Config
156. Model lineage and deployment approval with SageMaker Model Registry
157. Clean up

Quick Reference — AWS AI Services & Bedrock

Key services used across the labs

- Amazon Bedrock — API access to foundation models, plus knowledge bases (RAG), Guardrails and model evaluation. Used from Lab 06 onward.
- Amazon SageMaker — train, deploy and evaluate models; Ground Truth (labelling), Clarify (bias and SHAP explainability), model cards and the Model Registry.
- Amazon Q Business — managed generative-AI assistant grounded in your own documents (Lab 11).
- Managed AI services — Comprehend, Translate, Polly, Transcribe and Rekognition (Lab 03).
- Amazon Macie — discovers PII in Amazon S3 (Lab 24).
- AWS CloudTrail + AWS Config — audit trail and continuous compliance for AI API activity (Lab 25).

CLI commands that appear in the labs

- `aws sts get-caller-identity` — confirm the identity and account you are using.
- `aws bedrock list-foundation-models` — list foundation models available in the Region.
- `aws bedrock-runtime invoke-model` — call a model for inference.
- `aws bedrock-agent list-knowledge-bases` / `aws bedrock-agent-runtime retrieve` — RAG knowledge bases.
- `aws sagemaker list-endpoints` / `list-notebook-instances` — find billing resources to clean up.
- `aws opensearchserverless list-collections` — find vector stores left behind by knowledge bases.
- `aws qbusiness list-applications` — list Amazon Q Business applications.
- `aws macie2 get-macie-session` — check Macie status.
- `aws s3 cp` / `aws s3 rm` / `aws s3 rb` — move data and delete buckets during cleanup.

Consolidating What You Learned

The five exam domains map directly onto the five practical-assessment abilities. Check that you can do each unaided:

- A1 (Domain 1) — Explain AI, ML, deep learning and generative AI, classify data and learning types, and match a task to an AWS AI service or SageMaker.
- A2 (Domain 2) — Enable Bedrock, compare foundation models, tune inference parameters, and reason about tokens, embeddings, capabilities and limitations.
- A3 (Domain 3) — Engineer prompts, build and tune a RAG pipeline, decide on fine-tuning, and evaluate model performance.
- A4 (Domain 4) — Apply Guardrails, detect bias with Clarify, produce SHAP explanations, and document a model with a model card.
- A5 (Domain 5) — Secure an AI workload with IAM, KMS and VPC endpoints, protect data with Macie, and govern it with CloudTrail, Config and the Model Registry.

Re-work the labs from memory and then apply the same techniques to a workload in your own organisation — being able to produce the artefacts unaided is the best consolidation.

Glossary

Key AIF-C01 terms used across the labs.

Term	Meaning
Artificial intelligence (AI)	The broad field of building systems that perform tasks normally requiring human intelligence.
Machine learning (ML)	A subset of AI in which a model learns patterns from data rather than following hand-written rules.
Deep learning	ML using multi-layer neural networks; the basis of modern language and vision models.
Generative AI	AI that produces new content — text, images, code — from a foundation model.
Foundation model (FM)	A large model pre-trained on broad data that can be adapted to many downstream tasks.
Large language model (LLM)	A foundation model trained on text that predicts and generates language.
Amazon Bedrock	AWS managed service giving API access to foundation models, knowledge bases and guardrails.
Amazon SageMaker	AWS service for building, training, tuning and deploying machine learning models.
Amazon Q Business	A managed generative-AI assistant that answers questions grounded in your enterprise documents.
Token	The unit of text a model reads and generates; token counts drive cost and the context window.
Embedding	A numeric vector representing meaning, used to measure semantic similarity and power retrieval.
Vector store	A database of embeddings queried by similarity; the retrieval backbone of RAG.
RAG	Retrieval augmented generation — grounding a model's answer in documents retrieved at query time.
Prompt engineering	Designing prompts — instructions, context and examples — to steer a model's output.
Zero/one/few-shot	Prompting with no, one, or several worked examples of the task in the prompt.
Temperature / top-P	Inference parameters that control randomness and the token sampling pool of the output.
Fine-tuning	Customising a foundation model further on your own labelled data.
Hallucination	A fluent but factually wrong model output; mitigated by grounding and human review.
Bedrock Guardrails	Configurable content, denied-topic, word and PII filters applied to Bedrock model calls.
SageMaker Clarify	A tool that measures bias metrics and produces SHAP feature-attribution explanations.
Model card	A document recording a model's intended use, risk rating, training details and evaluation.
Responsible AI	Building AI that is fair, explainable, transparent,

	safe and privacy-respecting.
Shared responsibility model	AWS secures the cloud; the customer secures what they build and put in it.
IAM / KMS	AWS Identity and Access Management for least-privilege access; Key Management Service for encryption keys.
Amazon Macie	A service that discovers and reports sensitive data such as PII stored in Amazon S3.
AWS CloudTrail	Records AWS API calls, providing the audit trail for who did what and when.